

DECODE LABS VIRTUAL INTERNSHIP

DATA ANALYTICS - WEEK 1 PROJECT REPORT

Project Name: Data Cleaning and Preparation Report

Dataset Used: E-Commerce Orders Dataset

Submitted By: Himani

Date: June 29, 2026

1. Introduction

The primary goal of this project is to clean, structure, and prepare the raw E-Commerce Orders Dataset for accurate business analysis. Raw datasets often contain errors, duplicate entries, missing details, and inconsistent formatting. Performing thorough data cleaning ensures that any future dashboard or metric calculation represents true business performance without errors.

2. Data Cleaning Steps Performed

Step 2.1:

- Duplicate Record AuditAction: Checked the entire dataset for duplicate order entries based on unique identifiers (Order ID and Customer ID).
- Method: Used the Excel 'Remove Duplicates' utility tool.
- Result: No duplicate rows were found, confirming that the collection of order logs was structurally sound from the start.

The removal tool scanned all rows to check for duplicate entry patterns in order logs. The validation confirmation message box has been attached to prove data compliance.

(Refer to Figure 1 on the next page)

Online	Delivered	TRK29241424	8	SAVE10	Email
Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram
Gift Card	Returned	TRK96417362	6	SAVE10	Facebook
Cash	Shipped	TRK78809193	9	FREESHIP	Facebook
Gift Card	C				Email
Credit Card	S			AVE10	Google
Credit Card	R			INTER15	Instagram
Online	R			AVE10	Email
Debit Card	P			REESHIP	Referral
Credit Card	R			REESHIP	Facebook
Debit Card	Pending	TRK79186539	3	SAVE10	Instagram
Cash	Delivered	TRK54930938	3		Google
Credit Card	Pending	TRK13194259	3	FREESHIP	Email
Debit Card	Shipped	TRK11972116	3		Referral
Gift Card	Pending	TRK65191045	3	SAVE10	Facebook

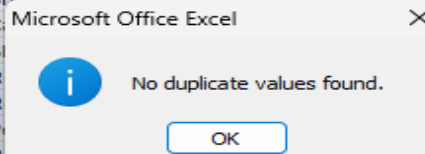


Figure 1: Excel Remove Duplicates tool execution confirming zero redundant rows.

Step 2.2:

- Handling MissingValues (Data Imputation)Action: Scanned all attributes for blank fields. Significant missing entries were discovered in the Coupon Code column.
- Method: Applied the 'Find and Replace' technique to standardize the data.
- Result: Replaced all 309 empty instances with the explicit text value "No Coupon" to ensure formatting uniformity and prevent analysis glitches.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
2	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.1
3	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	302.7
4	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2753.4
5	ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
6	ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04
7	ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram	491.72
8	ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook	664.42
9	ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FREESHIP	Facebook	747.75
10	ORD200008	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2	No Coupon	Email	268.56
11	ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10	Google	2037.52
12	ORD200010	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram	3129.85
13	ORD200011	2024-02-17	C93861	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234646	7	SAVE10	Email	147.42
14	ORD200012	2024-10-15	C38785	Monitor	2	180.5	179 Main St	Debit Card	Pending	TRK20419991	6	FREESHIP	Referral	361
15	ORD200013	2023-08-30	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FREESHIP	Facebook	604.47
16	ORD200014	2023-03-27	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending	TRK79186539	3	SAVE10	Instagram	786.66
17	ORD200015	2023-07-17	C39416	Printer	1	473.96	942 Main St	Cash	Delivered	TRK54930938	3	No Coupon	Google	473.96

Figure 2: Missing fields standardized by replacing blank cells with 'No Coupon'.

Step 2.3:

Data Validation & Quality ChecksAction: Reviewed operational metrics including Total Price calculations and the consistency of Date formats.

Method: Inspected formulas across samples to verify that Quantity multiplied by Unit Price correctly matched the stated Total Price. No mathematical or timezone formatting discrepancies were detected.

3. Data Cleaning Log (Audit Trail)

The table below documents the structured log created on a separate sheet (Sheet2) to track the exact changes made to the dataset for documentation purposes:

	A	B	C	D
1	Change ID	Column Description	Impact on Data	Status
2	CHG-01	Order ID & Customer ID	Checked for duplicate records using Remove Duplicates tool. No duplicate rows were found.	Verified
3	CHG-02	Coupon Code	Identified 309 blank cells and replaced them with 'No Coupon' for data consistency.	Completed
4	CHG-03	Total Price Column	Cross-checked the mathematical calculation (Quantity x Unit Price) across all rows.	Verified
5	CHG-04	Order Date	Reviewed date formats to ensure uniformity across the entire dataset.	Completed

Figure 3: Structured Data Cleaning Log documenting all modifications and data validation steps.

4. Conclusion

The E-Commerce Orders Dataset has been successfully audited, cleaned, and standardized. All empty fields have been populated with correct placeholders, and data redundancy risks have been eliminated. The structured file is now 100% prepared to be imported into visualization engines like Power BI or Excel Charts for Phase 2 analysis.